

| Req ID | Type       | Description                                                                                                                                                             | Priority | Acceptance Criteria                                                                                                                                                                                           | Rationale                                                                  |
|--------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| FR-001 | Functional | The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates, and trigger alerts at 30, 15, and 3 days before expiry. | High     | <b>Pass:</b> An alert is queued when a monitored certificate reaches 30, 15, or 3 days before expiry. <b>Fail:</b> An expiring certificate reaches one of these thresholds without an alert.                  | Prevents unexpected SSL/TLS certificate expiration and service disruption. |
| FR-002 | Functional | The system shall perform daily WHOIS registration checks for all monitored domains and record their registration expiration dates.                                      | High     | <b>Pass:</b> The system completes a WHOIS check for every monitored domain and stores its expiration date. <b>Fail:</b> A monitored domain has no updated WHOIS expiration information after the daily check. | Enables proactive detection of domain registration expiry.                 |

|               |            |                                                                                                                                                                     |               |                                                                                                                                                                                                                             |                                                                               |
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| <b>FR-003</b> | Functional | The system shall maintain a configurable list of domains and SSL endpoints to be monitored.                                                                         | <b>High</b>   | <b>Pass:</b> An authorized user can add, modify, or remove a domain/SSL endpoint, and the updated list is used during the next monitoring cycle. <b>Fail:</b> An authorized change is not reflected in the monitoring list. | Allows administrators to keep the monitoring scope current.                   |
| <b>FR-004</b> | Functional | The system shall notify the SysAdmin and Security Officer through an escalation ladder when a domain or certificate approaches its configured expiration threshold. | <b>High</b>   | <b>Pass:</b> Notifications are sent according to the configured escalation level at the applicable expiry threshold. <b>Fail:</b> A required notification is not generated or escalated.                                    | Ensures that unresolved expiration risks reach the appropriate personnel.     |
| <b>FR-005</b> | Functional | The system shall record the results of WHOIS and SSL/TLS audits, including the domain,                                                                              | <b>Medium</b> | <b>Pass:</b> Each completed audit produces a record containing all specified fields. <b>Fail:</b> Any completed audit lacks one or                                                                                          | Provides an audit trail for monitoring, troubleshooting, and security review. |

|                 |                                        |                                                                                                                                                   |             |                                                                                                                                                                                                                                                              |                                                                                                  |
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|                 |                                        | audit status, expiration date, and alert status.                                                                                                  |             | more required fields.                                                                                                                                                                                                                                        |                                                                                                  |
| <b>NF R-001</b> | Nonfunctional — Performance & Security | The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.                                                    | <b>High</b> | <b>Pass:</b> Benchmarking confirms that 1,000 domains and SSL endpoints are scanned in less than 3 minutes while maintaining the required security standards. <b>Fail:</b> The scan exceeds 3 minutes or fails the required security checks.                 | Ensures the monitoring system can operate efficiently at scale.                                  |
| <b>NF R-002</b> | Nonfunctional — Reliability            | The system shall complete the scheduled daily monitoring cycle without losing audit results when an individual domain or endpoint is unreachable. | <b>High</b> | <b>Pass:</b> During a test containing unreachable endpoints, the system records the failure and continues processing the remaining endpoints, with no loss of completed audit records. <b>Fail:</b> One unreachable endpoint terminates the monitoring cycle | Prevents a single network or endpoint failure from interrupting the entire monitoring operation. |

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|  |  |  |  | or causes<br>completed<br>results to be lost. |  |
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